Theorems:

Week 1: **Theorem 1.1** Let A and B be statements. Then A $\Rightarrow$ B is equivalent to $\overline{\text{A}}\vee$B. That is, (A$\Rightarrow$B) $\iff$ ($\overline{\text{A}}$ $\vee$B).

Theorem 1.2 Let A and B be statements. Then A $\Rightarrow$ B is equivalent to $\overline{\text{B}} \Rightarrow \overline{\text{A}}$. That is, (A $\Rightarrow$ B) $\iff$ ($\overline{\text{A}}\vee$B).

Theorem 1.3 Let A and B be statements. Then $\overline{\text{A} \vee \text{B}}$ is equivalent to $\overline{\text{A}} \wedge \overline{\text{B}}$.

Theorem 2.4 Given any two real numbers a < b, there is some r $\in$Q such that a <r < b.

Theorem 2.5 The number $\sqrt{2}$ is irrational.

Week 2:

Theorem 3.7 -AM-GM Inequality Let n$\in$N and $x_1, \ldots, x_n \geq 0$. Then $(x_1 \cdot x_2 \ldots x_n)^{1/n} \leq \frac{(x_1+\ldots+x_n)}{n}$

Theorem 3.8 -Cauchy-Schwarz Inequality Let n $\in$ N, $x_1,...,x_n$, y_1, ..., y_n $\in$ R. Then $x_1y_1 + \cdots + x_ny_n \leq \sqrt{x_1^2 + ... + x_n^2}\ \sqrt{y_1^2 + ... + y_n^2}$

Theorem 3.8 Triangle Inequality For x,y $\in$ R, $|x + y| \leq |x| + |y|$.

Corollary 3.12 Reverse Triangle Inequality $||x| - |y|| \leq |x + y|$. in particular: $|x| - |y| \leq |x + y|$

Theorem 4.1 -Geometric Series Let x $\neq$ 1 be a real number. Then for N $\in$ N, $\sum_{n=1}^{N} x^n = \frac{x - x^{N+1}}{1-x}$ and $\sum_{n=0}^{N} x^n = \frac{1-x^{N+1}}{1-x}$

Theorem 4.2 Every Integer n $\geq$ 2 can be written as a product of primes n = $p_1...p_k$

Week 3: **Completeness Axiom for the real numbers:** Every nonempty subset of R that is bounded above has a least upper bound.

Theorem 5.1 he Archimedean property holds. That is, for any x, y $\in$ R with x, y > 0, there is some n $\in$ N such that ny > x.

Theorem 6.6 Let x > 0 and k $\in$ N. Then there is a unique y > 0 such that y^k = x.

Week 4: **Theorem 7.1 Monotone Convergence Theorem** Let (a_n) be an increasing sequence of real numbers that is bounded above (i.e. there is M so that $a_n \leq$ M for all n). Then (a_n) converges to some limit. If (a_n) is a decreasing sequence and is bounded below, then (a_n) converges to some limit.

Theorem 8.1 Given a sequence (a_n) with $a_n \in$ 0, 1, . . . , 9 for all n, the decimal expansion x = $0.a_1a_2...a_n...$ given by the limit of the sequence $x_n = \frac{a_1}{10} + \frac{a_2}{100} + \ldots + \frac{a_n}{10^n}$ defines a real number x satisfying $0 \leq x \leq 1$

Theorem 8.2 Let x $\in$ R satisfy 0 $\leq$ x < 1. Then there is a sequence of integers (a_n), with $a_n \in$ 0, 1, . . . , 9 for all n, so that if we let $x_n = \frac{a_1}{10} + \frac{a_2}{100} + \ldots + \frac{a_n}{10^n}$ then $x_n \to$ x as n $\to \infty$

Theorem 8.3 Let 0 $\leq$ x < 1 and suppose that x = $0.a_1a_2... = 0.b_1b_2....$ Let ℓ be the smallest integer for which $a_\ell \neq b_\ell$ and suppose that $a_\ell < b_\ell$. Then $b_\ell = a_\ell + 1$, and for all k > ℓ we have b_k = 0 and a_k = 9

Lemma 8.4 if x $\geq$ 0 is rational, then it has a periodic decimal expansion, that is, x = $a_0.a_1.a_2 \ldots a_k\overline{b_1b_2 \ldots b_\ell}$

Corollary 8.5 If p and q are positive integers, the rational number p/q has a decimal expansion with period at most q.

Lemma 8.6 If x $\geq$ 0 has a periodic decimal expansion, then x is rational.

Theorem 8.7 A number x $\geq$ 0 is rational $\iff$ it has a periodic decimal expansion.

Theorem 8.11 -Comparison Test If $b_j \geq 0$ for all j and $\sum_{j=1}^{\infty} b_j$ is convergent, and if $|a_j| \leq b_j$ for all j, then $\sum_{j=1}^{\infty} a_j$ is also convergent, and moreover $|\sum_{j=1}^{\infty} a_j| \leq \sum_{j=1}^{\infty} b_j$

Week 5: **Lemma 9.1** For all u,v $\in$ C: (1) $\overline{u+v} = \bar{u} + \bar{v}$ (2) $\overline{uv} = \bar{u} * \bar{v}$ (3) $|u+v| = |u| + |v|$

Theorem 9.2 -De Moivre's If we let z = r(cosθ + isinθ), and n $\in$ N then $z^n = r^n(\cos n\theta + i\sin n\theta)$ $z^{-n} = r^{-n}(\cos -n\theta + i\sin -n\theta) = \frac{1}{r^{-n}}(\cos(n\theta) - i\sin(n\theta))$

Theorem 9.3 -Eulers Formula $e^{i\theta} = \cos(\theta) + i\sin(\theta)$.

Theorem 9.4 -Roots of Unity The solutions to z^n = 1 are 1, $w, \ldots w^{n-1}$ where $w = e^{\frac{2\pi i}{n}}$ That is, they are $e^{\frac{2\pi k i}{n}}$ for k = 0, 1, . . ., n - 1.

Theorem 10.1 -Abel-Ruffini There is no formula (like the quadratic formula) for the roots of a polynomial of degree $\geq$ 5

Theorem 10.2 -Fundamental Theorem of Algebra Any complex polynomial has at least one root in C.

Theorem 10.3 -Factorization Theorem If p is a degree n polynomial, then there are n roots $r_1, \ldots, r_n \in$ C and a number a $\in$ C so that $p(z) = a(z - r_1)(z - r_2) \ldots (z - r_n)$.

Theorem 10.4 -Real Polynomials have conjugate roots If p(x) has *real* coefficients and r is a root, so is $\bar{r}$.

Theorem 10.5 -Root-Coefficient Theorem $p(x) = x^n + a_{n-1}x^{n-1} + \ldots + a_1x + a_0$, has roots $r_1, \ldots, r_n$ (counting multiplicities), then $r_1 + \ldots + r_n = -a_{n-1}$ $r_1 \ldots r_n = (-1)^n a_0$. In general, if s_j denotes the sum of all products of j-tuples of the roots (e.g. $s_2 = r_1r_2 + r_1r_3 + r_2r_3 + \ldots$), then $s_j = (-1)^j a_{n-j}$.

Week 6: **Theorem 11.1 - The Remainder Theorem** Let a $\in$ N and b $\in$ Z. There are unique integers $q \in$ Z and 0 $\leq$r<a such that b = qa+r

Lemma 11.2 If a divides b and b divides a then a = $\pm$b

Lemma 11.3 -Linear Combination Lemma Let a, b, and d be integers. If d divides both a and b, then d divides every integer linear combination of a and b. In other words, for every pair of integers m, n $\in$ Z, the integer d also divides ma + nb.

Theorem 11.4 -Bezout's Identity Let a and b be integers. Then gcd(a, b) is an integer linear combination of a and b. That is, there are integers s and t such that gcd(a, b) = sa + tb.

Corollary 11.5 Let a, b, c $\in$ Z. Suppose c $\neq$ 0, c divides ab and gcd(a, c) = 1. Then c divides b.

Corollary 11.6 Let a, b $\in$ Z and let d $\in$ Z be such that d divides a and d divides b. Then d divides gcd(a, b).

Corollary 11.7 Let d be a common divisor of a and b, which is divisible by all divisors of a and b. Then d = $\pm$gcd(a, b).

Corollary 11.8 If a, b $\in$ Z, p is prime, and p divides ab, then either p divides a or p divides b (or both).

Corollary 11.9 If n = $p_1^{m_1} \ldots p_k^{m_k}$, where each of $p_1, \ldots, p_k$ is prime, and if p is a prime number that divides n, then p = p_i for some i = 1, . . . , k.

Theorem 11.10 -Fundamental Theorem of Arithmetic Let n ≥ 2 be an integer.*Existence* Then n is equal to a product $n = p_1^{r_1} \ldots p_k^{r_k}$ of powers of prime numbers, where $p_1 < \ldots < p_k$ and r_i ¿ 0 for all i. (Uniqueness) The factorization is unique. That is, if we also have $p_1^{r_1} \ldots p_k^{r_k} = n = q_1^{s_1} \ldots q_\ell^{s_\ell}$ where $q_1 < \ldots < q_\ell$ are primes and $s_i > 0$, then $k = \ell$, $p_i = q_i$, and $r_i = s_i$ for all i.

Lemma 11.11 -Existence of prime decomposition, Part 1. Every integer $n \geq 2$ can be written as a product of primes $n = p_1 \ldots p_k$.

Lemma 11.12 -Existence of prime decomposition, Part 2. Every integer $n \geq 2$ can be written as a product of powers of primes $n = p_1^{r_1} \ldots p_k^{r_k}$ where $p_1 < \ldots < p_k$ and $r_i \geq 0$.

Lemma 11.13 -Uniqueness of Prime decomposition. Every integer n can be written as a *unique* product of powers of primes $n = p_1^{r_1} \ldots p_k^{r_k}$

Theorem 11.14 Let $n = p_1^{a_1} \ldots p_k^{a_k}$ be a prime decomposition. Then m divides n if an only if $m = p_1^{b_1} \ldots p_k^{b_k}$ with each $0 \leq b_i \leq a_i$

Theorem 11.15 Let a and b have prime factorizations, $a=p_1^{r_1} \ldots p_m^{r_m}$, $b = p_1^{s_1} \ldots p_m^{s_m}$ Here p_i's are distinct, but r_i and s_i are allowed to be zero. Then: (1) gcd(a,b) = $p_1^{min(r_1,s_1)} \ldots p_m^{min(r_m,s_m)}$ (2) lcm(a,b) = $p_1^{max(r_1,s_1)} \ldots p_m^{max(r_m,s_m)}$ (3) lcm(a,b) = ab/gcd(a,b).

Lemma 11.16 Let $r \in Q$. Then there exist unique coprime integers a and b such that $r = \frac{a}{b}$ and $b \geq 1$.

Theorem 11.17 Let n be a positive integer. Then $\sqrt{n} \in Q$ if and only if n is a perfect square, that is, $n = m^2$ for some integer m.

Theorem 11.18 If a, b $\in$ N are coprime, and ab is an nth power, then so are a and b.

Week 7: **Theorem 12.1** Let $m \in N$ and a,b $\in$ Z. The following are equivalent: (1) a =b mod m; (2) m divides a - b; (3) a and b have same remainder when divided by m; (4) b = qm + a for some $q \in Z$.

Theorem 12.2 We have the following properties for integers a, b, c $\in$ Z and $m \in N$: (1) a = a mod m;(2) If a = b mod m, then b = a mod m. (3) If a = b mod m and b = c mod m, then a = c mod m.

Theorem 12.3 Suppose a = x mod m and b = y mod m. Then (1) -a = -x mod m; (2) a + b = x + y mod m; (3) ab = xy mod m; (4) or any natural number k, one has $a^k = x^k$ mod m.

Theorem 12.4 Let a and m be coprime. If x, y $\in$ Z and xa = ya mod m, then x = y mod m. In particular, if p is prime and p does not divide a, then xa = ya mod p implies that x = y mod p.

Theorem 12.5 The equation ax = b mod m has a solution if and only if gcd(a, m) divides b.

Theorem 12.6 Let $m \geq 2$ be a natural number. Then Z/m is a field if and only if m is prime.

Theorem 12.7 -Fermat's Little Theorem Let a be an integer, and let p be a prime number. If p does not divide a, then $a^{p-1} - 1$ mod p

Theorem 12.8 -Fermat's Little Theorem Let $n \in N$ and p be a prime number. Assume n and p - 1 are coprime and p does not divide b. Then the equation $x^n = b$ mod p. has exactly one solution in $x \in 1, \ldots, p-1$

Week 8: **Theorem 13.1** Let $f : X \to Y$ and $g : Y \to Z$ be functions. (1) If f and g are injective, so is $g \circ f$. (2) If f and g are surjective, so is $g \circ f$. (3) If f and g are bijective, so is $g \circ f$.

Theorem 13.2 Let $f : X \to Y$ be a function and B, $B' \subseteq Y$. Then $f^{-1}(B \cap B') = f^{-1}(B) \cap f^{-1}(B')$ and $f^{-1}(B \cup B') = f^{-1}(B) \cup f^{-1}(B')$

Theorem 13.3 Let X and Y be finite sets and let $f : X \to Y$ be a function. (1) If f is injective, then $|X| \leq |Y|$. (2) If f is surjective, then $|X| \geq |Y|$. (3) If f is bijective, then $|X| = |Y|$.

Theorem 14.1 $Q \cong N$.

Theorem 14.2 Let S be a set and let P(S) denote its power set, consisting of the subsets of S, i.e. P(S) = A : A $\subseteq$ S. Then $|S| < |P(S)|$.

Week 9: **Theorem 15.1 -Multiplication Principle** Let P be a process consisting of n stages such that at each stage i, there are a_i choices we can make, and no two distinct choices can result in the same outcome. Then after n stages, the total number of outcomes is $a_1 a_2 \ldots a_n$.

Theorem 15.3 -Ordering Theorem Given a set S of n distinct objects, there are n! ways of ordering them.

Theorem 15.5 For integers $0 \leq k \leq n$, we have $\begin{pmatrix} n \\ k \end{pmatrix} = \frac{n!}{k!(n-k)!}$

Corollary 15.6 For $n \geq 0$, we have $2^n = \sum_{k=0}^{n} \begin{pmatrix} n \\ k \end{pmatrix}$

Theorem 15.8 For $n \in N$, $k \geq 2$ and nonnegative integers $r_1, \ldots, r_k$ such that $r_1 + \ldots + r_k = n$, we have $\begin{pmatrix} n \\ r_1 \ldots, r_k \end{pmatrix} = \frac{n!}{r_1! r_2! \ldots r_k!}$

Theorem 15.9 -The Binomial Theorem Let $n \in N$ and a, b $\in$ C. Then $(a+b)^n = \sum_{k=0}^{n} \begin{pmatrix} n \\ k \end{pmatrix} a^k b^{n-k}$

Theorem 15.10 -The Multinomial Theorem Let $n \in N$ and $X_1, \ldots, x_k \in C$. Then $(x_1 + \ldots + x_k)^n = \sum \begin{pmatrix} n \\ r_1 \ldots, r_k \end{pmatrix} x_1^{r_1} \ldots x_k^{r_k}$ where the sum is taken over all $r_i \in$ 0,1,. . .,n such that $r_1 + \ldots + r_k$ =n

Theorem 15.11 -Inclusion Exclusion Principle Let $n \geq 2$ and suppose $A_1, A_2 \ldots, A_n$ are finite sets Let $N_1 = |A_1| + |A_2| + \ldots + |A_n|$, $N_2 = |A_1 \cap A_2| + |A_1 \cap A_3| + |A_2 \cap A_3| + \ldots + |A_{n-1} \cap A_n|$, and similarly, let $N_k = \begin{pmatrix} \Sigma \\ 1 \leq i_1 < i_2 < \ldots < i_k \leq n \end{pmatrix}$ $|A_{i_1} \cap A_{i_2} \cap \ldots \cap A_{i_k}|$ be the sum of the sizes of the intersections of all collections of k different A_i's, so that $N_n = |\bigcap_{k=1}^{n} A_k|$

Then it equals $\sum_{k=1}^{n}(-1)^{k+1}N_k$. In particular, for finite sets A,B,C $|A \cup B| = |A| + |B| - |A \cap B|$, and $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$.

Week 10: **Lemma 16.1** The set S_n consists of exactly n! permutations.

Lemma 16.2 The set S_n equipped with the composition rule $\circ$ has the following properties (1)Closure: for any f and g in S_n, one has $f \circ g \in S_n$, (2)Associativity: for any f, g, and h in S_n, one has $f \circ (g \circ h) = (f \circ g) \circ h$, (3)Identity element: there is unique permutation $\iota \in S_n$ such that $f \circ \iota = \iota \circ f = f$ for any $f \in S_n$ (and we call ι the identity permutation), (4)Inverse: for any $f \in S_n$, there is unique $f^{-1} \in S_n$ such that $f \circ f^{-1} = f^{-1} \circ f = \iota$.

Lemma 16.4 If $a_1, \ldots, a_r$ are distinct numbers in 1, 2, . . . , n and $f = (a_1\ a_2 \ldots a_r)$ is a permutation, then the order of f is r.

Lemma 16.5 Let $f \in S_n$ have order m. If $f^k = \iota$ for some integer $k \in N$, then m divides k.

Theorem 16.6 Any permutation $f \in S_n$ is a composite of disjoint cycles, that is, $f = f_1 \ldots f_k$ for some $k \geq 1$ and disjoint cycles $f_1, \ldots, f_k$.

Lemma 16.7 Let $f \in S_n$ and let m be its order. Write $f = \sigma_1\sigma_2 \ldots \sigma_s$ where $\sigma_1, \ldots, \sigma_s$ are disjoint cycles of length $r_1, \ldots, r_s$ respectivly. Then m=lcm($r_1, r_2, \ldots, r_s$), where lcm is the smallest positive integer that is divisible by r_i for each i=1,. . .,s.

Lemma 16.8 Every permutation in S_n is a product of cycles of length 2.

Lemma 16.9 Let $n \in N$ (a) The sign of ι is 1 and any 2-cycle is -1 (b)or any f and g in S_n, we have sgn(fg) = sgn(f)sgn(g). (c) the sign of a cycle of length r is $(-1)r^{-1}$ (d)For every permutation $f \in S_n$, let $f = \sigma_1\sigma_2 \ldots \sigma_s$ where $\sigma_1, \ldots, \sigma_s$ are disjoint cycles of lengths $r_1, \ldots, r_s$.Then sgn(f) = $(-1)^{r_1-1}(-1)^{r_2-1}(-1)^{r_3-1} \ldots (-1)^{r_s-1}$ (e) If $f \in S_n$, then sgn(f) = sgn(f^{-1})

Corollary 16.10 Let f be a permutation in S_n (1)f is even if and only if it is a product of even number of cycles of order 2. (2)f is odd if and only if it is a product of odd number of cycles of order 2.